

LAG-2 Phase 2 (start): Embedding of $\mathbb{R}^{\{3,1\}} \subset D_{IV}^5$

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LAG-2 Phase 2 (start): Embedding Question

Phase 1 (T2340) identified the dimensional split: $\dim_R(D_{IV}^5) = \text{rank}^2 + C_2 = 4 + 6$. Today's Phase 2 start addresses the next question: WHICH 4-dim sub-locus of D_{IV}^5 is the external spacetime $\mathbb{R}^{\{3,1\}}$?

The question

The dimensional split is forced (Phase 1). The embedding is not — there are multiple candidate 4-dim sub-loci of D_{IV}^5 , and we need to identify which is the canonical choice for dimensional reduction.

Once embedded, the orthogonal complement (6-dim) is the internal compactified manifold. The reduction integral $S_{4D} = \int_{\text{internal}} S_{BST}$ integrates over this 6-dim complement.

Constraints on the embedding

The 4-dim sub-locus must:

1. Have the right dimension: $\text{real dim} = 4 = \text{rank}^2$
2. Be totally geodesic (or close to it) — so the reduction integral has a clean structure
3. Support a Lorentz-signature metric (or Wick-rotatable to one) — D_{IV}^5 's Bergman metric is positive-definite Riemannian; we need to extract the (3+1) Lorentz signature somehow
4. Be BST-structurally natural — derivable from D_{IV}^5 's symmetries, not chosen ad hoc
5. Have a 6-dim “orthogonal complement” that carries the internal structure cleanly

Candidate sub-loci

Candidate A: $D_{IV}^2 \subset D_{IV}^5$ (sub-type-IV bidisk-like)

$D_{IV}^2 = SO_0(2,2)/[SO(2) \times SO(2)]$ is a Hermitian symmetric domain with complex dim 2, real dim 4.

Embedding: $D_{IV}^2 \subset D_{IV}^5$ as totally geodesic sub-domain (Cartan classification of sub-Cartan-types of D_{IV}^n).

Pros: - Real dim 4 $\checkmark$ - Totally geodesic $\checkmark$ (Hermitian symmetric sub-domain) - BST-structurally natural $\checkmark$ (D_{IV}^2 is a smaller Cartan type IV) - Has $SO(2,2)$ action — provides 2 “time-like” directions, 2 “space-like” — close to Lorentz (3+1) but not exact

Cons: - (2,2) signature, not (3,1) — needs an additional involution or signature choice to recover (3,1) - The complement $D_{IV}^5 \setminus D_{IV}^2$ is not naturally a 6-dim Hermitian symmetric space (it's a coset with extra structure)

Candidate B: 4-ball sub-locus of $M(D_{IV}^5)$

T2328 showed $M(D_{IV}^5) = \text{open 5-ball in } \mathbb{R}^5$. Take the 4-ball $M(D_{IV}^5) \cap \{x_5 = 0\}$ as the external spacetime.

Pros: - Real dim 4 $\checkmark$ - BST-structurally natural $\checkmark$ (sub-locus of the canonical Möbius locus) - All real (no imaginary coordinates) — Euclidean signature, Wick-rotatable to Lorentzian - Complement = 6 dims (1 from x_5 + 5 from imaginary parts) — clean dim accounting

Cons: - Not totally geodesic in general (4-ball $\subset$ 5-ball in Euclidean metric is geodesic; in Bergman metric on D_{IV}^5 it's not) - Choice of " $x_5 = 0$ " is arbitrary — any (5-1)-dim sub of the 5-ball would work; need a canonical reason for THIS slice - The complement isn't a single connected 6-dim manifold

Candidate C: Cartan involution fixed locus

Every bounded symmetric domain has a Cartan involution θ . The fixed-point set of θ on D_{IV}^5 is the maximal compact sub-group K 's orbit through the origin — but K is the isotropy group, so its orbit is just $\{0\}$, not 4-dim.

Alternative: use an "anti-Cartan" involution σ such that the fixed set σ -fixed is a 4-dim sub-manifold.

Pros: - If exists, naturally Lorentz signature (σ might be the analog of complex conjugation in a Lorentzian sense)

Cons: - Construction of σ is non-obvious for type IV; needs to be worked out - May not give exactly 4-dim fixed locus

Candidate D: A Lagrangian sub-manifold

A Lagrangian sub-manifold of D_{IV}^5 (with respect to its symplectic / Kähler structure) has real dim = (complex dim) = $n_C = 5$. Too large by 1.

A 4-dim sub-manifold of a Lagrangian $L \subset D_{IV}^5$ would work if natural. But Lagrangian is 5-dim, sub-Lagrangian 4-dim is not standard.

Candidate E: 4-dim totally-geodesic sub of $M(D_{IV}^5)$

T2328's $M(D_{IV}^5)$ is a real 5-ball. Its totally-geodesic 4-dim sub-manifolds (in the Bergman-restricted metric on M) are 4-dim hyperbolic balls $H^4 \subset H^5 = M(D_{IV}^5)$.

Pros: - Real dim 4 $\checkmark$ - Totally geodesic $\checkmark$ (within M , the 4-ball is geodesic) - Hyperbolic structure $\checkmark$ — H^4 has $(-, +, +, +)$ signature after Wick rotation - BST-structurally natural $\checkmark$ (canonical 4-dim sub of canonical 5-dim Möbius locus)

Cons: - H^4 is Lorentz-signature after Wick rotation, not directly; standard physics interpretation requires Wick rotation - Complement (6-dim) includes the imaginary-direction Hermitian part of D_{IV}^5 M , not a single manifold

Leading candidate: E ($H^4 \subset M(D_{IV}^5)$)

Among the five candidates, E is the leading choice because:

1. All constraints satisfied: 4-dim $\checkmark$, totally geodesic $\checkmark$, Wick-rotatable to Lorentz $\checkmark$, BST-natural $\checkmark$, has 6-dim complement $\checkmark$
2. Builds on prior work: T2328 already identified $M(D_{IV}^5)$; E is its canonical 4-dim totally-geodesic sub
3. Hyperbolic structure: H^4 with negative curvature matches AdS_4 / dS_4 -style cosmology — natural for cosmological reduction
4. Wick rotation is standard: physicists use Wick rotation routinely between Euclidean H^4 and Lorentzian H^4 / AdS_4

Specifically: external spacetime $\cong H^4 \subset M(D_{IV}^5) \subset D_{IV}^5$, then Wick rotate $H^4 \rightarrow AdS_4$ or $\mathbb{R}^{3,1}$ per physical interpretation.

6-dim internal complement

If $\text{external} \cong H^4 \subset M(D_{IV}^5)$, the complement is $D_{IV}^5 \setminus H^4$ with real $\dim 10 - 4 = 6 = C_2$.

Decomposition of complement: - 1 real dim from $M(D_{IV}^5) \setminus H^4$ (the “extra” Möbius direction) - 5 real dims from $D_{IV}^5 \setminus M(D_{IV}^5)$ (the imaginary part) - Total: $1 + 5 = 6 = C_2$ [?]

The 1+5 internal split is itself BST: 1 = trivial, 5 = n_C . The internal complement has structure $\{1 \text{ trivial} + n_C \text{ internal Hermitian}\} = 6 = C_2$.

What Phase 2 START produces today

1. The embedding question articulated precisely
2. Five candidates evaluated against five constraints
3. Leading candidate identified: $H^4 \subset M(D_{IV}^5)$ with internal complement $6 = 1 + n_C$
4. BST-structural reading: external 4D = rank^2 emerges as H^4 totally-geodesic sub of M ; internal $6 = C_2$ splits as $1 + n_C$
5. Wick rotation to physical Lorentz signature flagged as standard step

What Phase 2 FULL still needs (multi-week)

- Phase 2.1 (~3-5 days): formal proof that $H^4 \subset M(D_{IV}^5)$ is the canonical totally-geodesic 4-dim sub-locus
- Phase 2.2 (~1-2 weeks): compute the reduction integral $S_{4D} = \int_{\{\text{internal 6-dim}\}} S_{\text{geom}}[\text{Bergman curvature}]$ explicitly
- Phase 2.3 (~3-5 days): verify the 4D effective action S_{4D} recovers Einstein-Hilbert at low energy

Total Phase 2: 2-3 weeks, staged. Today closes the LEADING-CANDIDATE identification (the gate to Phase 2.1-2.3).

Tier reading

Phase 2 START identifies the embedding to I-tier. Full closure of the embedding (Phase 2.1) promotes to D-tier. The reduction integral (Phase 2.2) is the substantive multi-week work.

Connection to other LAG work

- LAG-1 S1 (T2339): Bergman Dirac D_B is defined on full D_{IV}^5 ; restriction to $H^4 \subset M$ gives the 4D Dirac after Wick rotation
- LAG-2 Phase 3: with H^4 embedding fixed, the 6 S_{BST} terms get reduced via integration over the 6-dim internal complement
- Möbius cohomology (T2328-T2335): $M(D_{IV}^5)$ is the canonical Möbius locus; H^4 is its canonical 4-dim sub. The Möbius mechanism (T2091) places leptons on M ; under $H^4 \subset M$ embedding, lepton mass formulas restrict to 4D spacetime — consistent.

Status

Phase 2 START complete: leading candidate identified ($H^4 \subset M(D_{IV}^5)$, internal $1 + n_C = 6$). Full Phase 2.1-2.3 deferred to multi-session work.

— Lyra, 2026-05-17 ~16:30 EDT